

Business Insights Report – Olist Customer Sentiment & Delivery Analysis

Executive Summary

The analysis highlights that delivery performance is the most influential driver of customer sentiment. Longer delays strongly correlate with negative reviews, while on-time or early deliveries significantly boost positive sentiment. Regional differences and product category expectations also play critical roles in shaping customer satisfaction.

Key Findings

1. Customer Experience & Sentiment

- Delivery delay is the most significant factor influencing customer sentiment.
- Early or on-time deliveries are associated with higher review scores.
- Order status categories (delivered, shipped, processing) show distinct sentiment patterns.

2. Regional Patterns

- Negative-review shares vary by region, reflecting different logistics performance levels.
- Top-rated product categories differ by region, suggesting localized customer expectations.

3. Product & Price/Cost Signals

- Product category and freight cost significantly influence satisfaction.
- Price and perceived value also shape sentiment, especially when coupled with delivery delays.

4. Model Performance & Explainability

- Multi-class model predicts Negative, Neutral, and Positive sentiments with ~80% accuracy.
- SHAP analysis confirms delivery delay, freight value, product category, and region as key drivers.
- Region label acts as contextual insight rather than a causal driver.

Actionable Recommendations

A) Operations & Logistics:

1. Set region-specific SLA buffers to reduce negative reviews.
2. Implement proactive delay communication and recovery protocols.
3. Review carrier and route performance by region.

B) Customer Experience:

4. Display dynamic delivery windows based on regional data.
5. Offer apology vouchers or incentives for delayed deliveries.

C) Merchandising & Pricing:

6. Localize category promotion by region.
7. Test free shipping thresholds and bundling offers.

D) Model & Data Improvements:

8. Add delivery-promise, distance, and carrier features.
9. Integrate text sentiment to refine Neutral/Positive distinction.
10. Monitor SHAP importance over time to detect data drift.

Quick Wins (Next 30 Days)

- Launch proactive late-order communication for predicted delays over 2 days.
- Adjust delivery windows in underperforming regions.
- Feature top-rated regional categories on homepages.
- Test free-shipping thresholds where freight impacts sentiment most.

KPIs to Track

- Negative review rate (overall & by region).
- On-time delivery rate.
- Average delivery delay (days).
- NPS proxy (Positive – Negative share).
- A/B test metrics for PDP messaging and freight promotions.
- SHAP top feature stability across retrains.

Appendix: Analysis Drivers

EDA showed strong correlation between delivery delay and sentiment, with regional variation in negative review rates and category preferences. Model explainability (SHAP) confirmed delivery delay days, freight value, price, product category, and region as key sentiment determinants. The model performs best on Positive vs Negative distinction, suggesting improvement potential in Neutral differentiation.